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Executive Summary Project

ColonyAI is an AI-powered Automated Plate Count Reader designed to modernize Total Plate Count (TPC) testing in microbiology laboratories. Currently, manual colony counting is time-consuming, inconsistent, and highly operator-dependent, with inter-analyst variability reaching up to 80%. Our solution addresses this by integrating a fine-tuned YOLOv8 computer vision model with a secure Next.js dashboard to automate plate localization, 5-class object detection (colony_single, colony_merged, bubble, dust_debris, media_crack), and CFU/ml calculation based on ISO/IEC Guide 98-3:2008.

By reducing analysis time by 85% and providing SHA-256 chained immutable audit trails, ColonyAI directly supports the Hackathon's focus on digital innovation and operational efficiency. For laboratory providers, this solution offers a transformative impact by eliminating human error, ensuring strict BPOM/ISO 17025 compliance, and enhancing diagnostic traceability, ultimately shifting laboratory operations toward high-value analytical work and reliable, audit-ready data.



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Problem Statement

Selected Case Statement

Case 1 — Microbiology Laboratory: Automated Plate Count Reader

Selected Sub-Case Statement

Automated detection, counting, and CFU/ml reporting of bacterial colonies from agar plate images, with classification across 5 object classes (colony_single, colony_merged, bubble, dust_debris, media_crack) and differentiation between valid colonies and artifacts, integrated into a web-based laboratory dashboard with enterprise-grade security and multi-tenant RBAC.

Main Objectives

The primary objective of ColonyAI is to replace subjective, manual colony counting with a standardized, automated digital pipeline. Our specific targets are:

- **Detection Accuracy:** Utilize a fine-tuned YOLOv8 model to identify 5 distinct object classes (colony_single, colony_merged, bubble, dust_debris, media_crack) with $\geq 92\%$ accuracy.
- **Operational Efficiency:** Reduce the Total Plate Count (TPC) analysis time from 15–30 minutes to under 2 minutes per sample.
- **Metrological Rigor:** Deliver reliable CFU/ml results by applying SA-001 area-based estimation for merged colonies and calculating GUM-compliant measurement uncertainty per ISO/IEC Guide 98-3:2008.
- **Traceability & Compliance:** Provide immutable, SHA-256 chained digital audit trails to support ISO 17025 accreditation and BPOM regulatory reporting requirements.
- **Data Security:** Implement enterprise-grade protections, including magic-bytes file validation, EXIF stripping, malware scanning, and granular 5-role RBAC to safeguard laboratory data.
- **Institutional Trust:** Provide a built-in Simulator module, enabling laboratory managers to compare AI counts against manual tallies to validate system performance before full adoption.



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Problem Definition

1.1 What is the main problem?

In Indonesian microbiology laboratories, Total Plate Count (TPC) remains the gold standard for testing microbial contamination. However, the process is entirely manual—analysts physically count colonies using a tally device under magnification. This reliance on human intervention creates three critical operational failures:

1. **High Inconsistency:** Manual counting is inherently subjective. Inter-analyst variability in counting the same sample routinely ranges from 22.7%–80% (ASTM F2944), leading to unreliable quality assessments.
2. **Throughput Bottlenecks:** Manual analysis is slow, with a single analyst limited to 20–40 plates/hour. This causes severe backlogs during peak periods, such as holiday food inspections and disease outbreak investigations.
3. **Critical Skill Dependency:** Accurate results depend on the analyst's experience to distinguish valid colonies from artifacts like air bubbles, dust debris, and media cracks. Junior analysts or under-resourced labs often struggle with this, risking the release of contaminated products.

1.2 Who is impacted and to what scale?

The impact of manual TPC testing affects the entire food safety ecosystem:

- **Food Manufacturers:** FMCG and dairy companies rely on rapid, accurate TPC results for production release decisions. Inaccurate results risk costly product recalls.
- **Regulatory Bodies (BPOM, Dinas Kesehatan):** Require standardized, auditable microbial records for certification and public safety enforcement.
- **Third-Party Testing Laboratories (KAN-accredited):** Face immense pressure to process increasing sample volumes with limited analyst capacity, directly impacting turnaround times and service quality.
- **Consumers:** Ultimately bear the health risk of foodborne illness if contaminated products reach the market due to testing inconsistencies.



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1.3 Prove the problem

The following evidence confirms that manual colony counting is a systemic risk to quality and public health:

Source	Key Finding
ASTM F2944	Manual counting shows inter-observer coefficients of variation between 22.7%–80%.
FDA BAM (2023)	Countable-range plates yield high variation, particularly for merged or irregular colonies.
BPOM Indonesia	18% of food violations involve microbiological non-conformance, often linked to inconsistent methods.
Lab Industry Survey	Colony counting consumes 40–60% of analyst working hours in TPC workflows.

These data points demonstrate that manual counting is not merely an operational bottleneck; it is a critical quality failure. The high coefficient of variation in manual counting invalidates reproducibility, while the labor-intensive nature of the process limits the volume of testing a laboratory can handle. These systemic risks directly impede the ability of Indonesian laboratories to meet increasing food safety demands, making an automated, standardized AI solution like ColonyAI a necessity for modern laboratory practice.



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Problem Solution

2.1 Main Solution

ColonyAI is a web-based intelligent laboratory platform that transforms agar plate images into accurate, standardized CFU/ml reports in under two minutes. The system integrates three tightly coupled components:

- **AI Vision Engine:** A fine-tuned YOLOv8 model trained to detect plate boundaries and classify objects into 5 categories: colony_single, colony_merged, bubble, dust_debris, and media_crack. Per-media-type confidence thresholds ensure high precision, while an SA-001 area-based estimation engine calculates counts within merged regions.
- **Intelligent Web Dashboard:** A secure, intuitive interface (Next.js 14) where analysts upload images and view color-coded annotated results. It handles all data entry, provides GUM-compliant measurement uncertainty, and enforces a mandatory digital sign-off process to ensure data integrity and accountability.
- **Simulator & Reporting Module:** A benchmarking tool allowing laboratory managers to compare AI counting accuracy against manual counts. Final results are exportable as audit-ready PDF (BPOM-compliant) and CSV reports.

2.2 Attachment Image / Illustration / Diagram

Figure 1: ColonyAI System Architecture and Data Flow Pipeline



Figure 2: ColonyAI Interface Dashboard

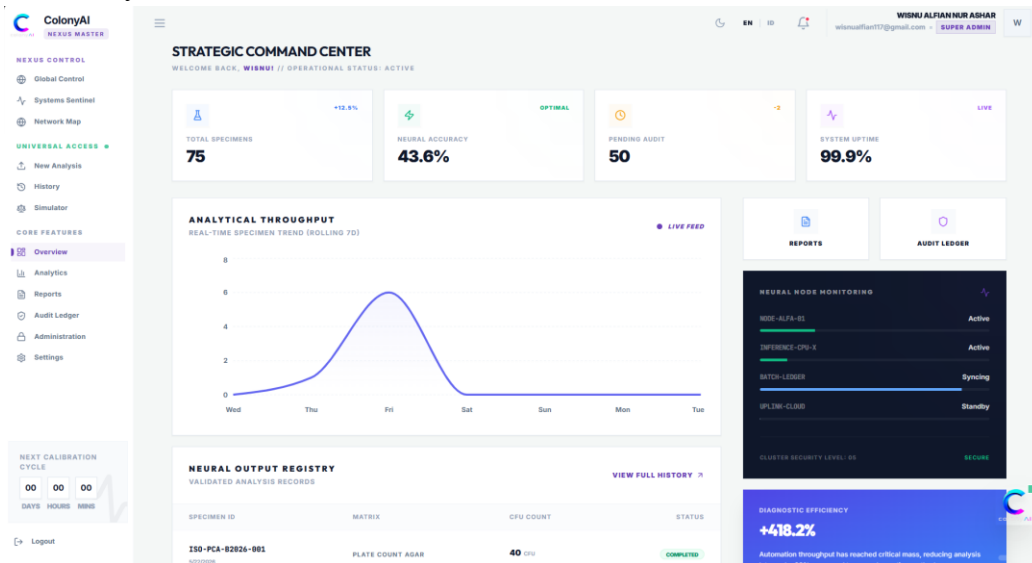


Figure 3: AI Inference & 5-Class Detection Result

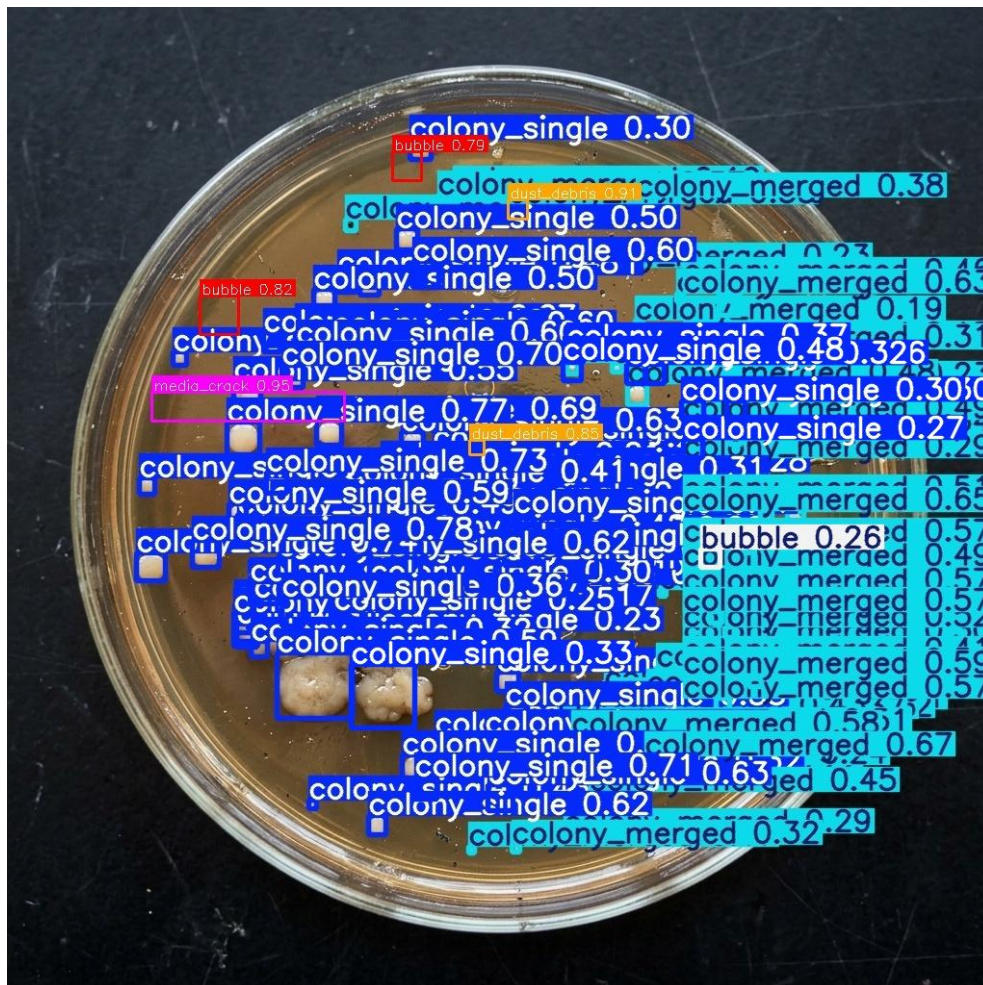


Figure 4: Automated CFU/ml Reporting & GUM Uncertainty

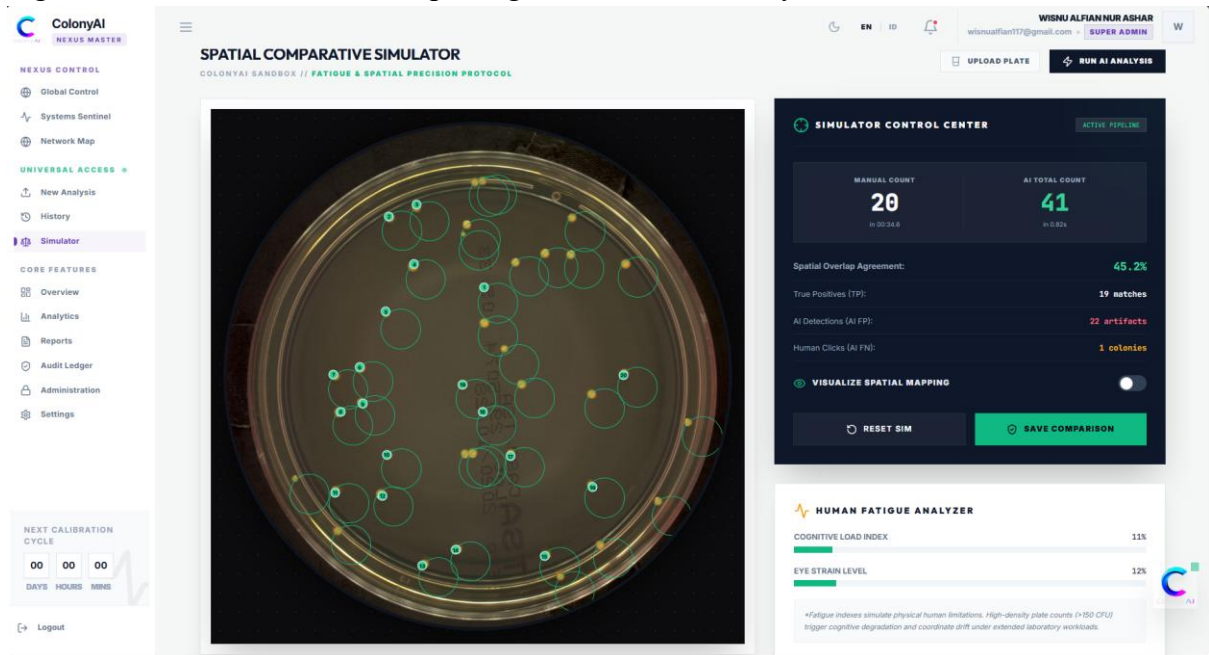
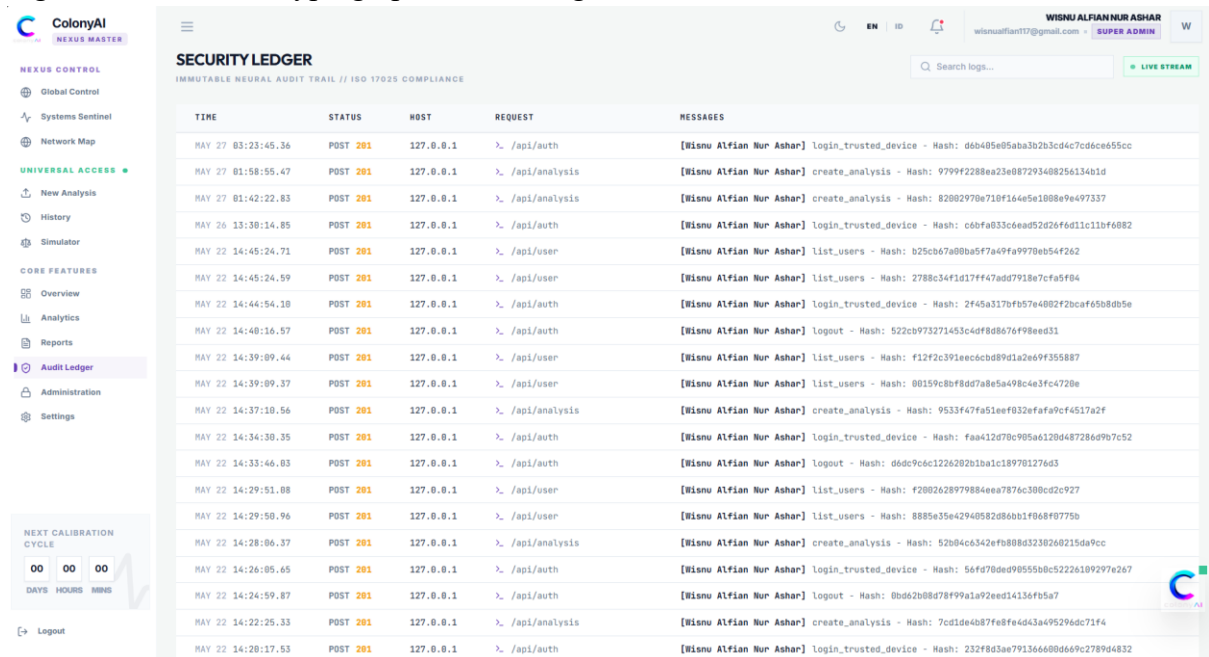


Figure 5: SHA-256 Cryptographic Audit Log





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2.3 How does the solution work?

The ColonyAI platform operates through a rigorous, automated pipeline:

- **Input:** Analysts upload agar plate images through the web dashboard, providing sample metadata (Sample ID, dilution factor, and media type).
- **Security & Pre-processing:** Before inference, files undergo multi-layer security: magic-bytes validation, malware scanning, and EXIF metadata stripping. The image is then normalized via OpenCV (CLAHE for contrast, Hough Circle for plate boundary detection, and homography for perspective correction).
- **AI Inference:** A fine-tuned YOLOv8 model detects objects, classifying them into five categories (colony_single, colony_merged, bubble, dust_debris, media_crack). An SA-001 engine estimates counts in merged regions.
- **Calculation & Governance:** The system computes CFU/ml using ISO 4833-1:2013 standards, applying GUM measurement uncertainty ($k=2$) for metrological traceability. Every result is recorded in an immutable, SHA-256-chained audit log in PostgreSQL.
- **Output:** The final dashboard displays the annotated results with color-coded bounding boxes. Analysts perform a mandatory digital sign-off, enabling the generation of BPOM-compliant PDF reports and direct LIMS integration.



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Impact & Outcome

3.1 Key Benefits of Adopting the Solution

ColonyAI delivers transformative benefits for microbiology laboratories, stakeholders, and regulatory compliance:

- **Operational Efficiency:** Reduces Total Plate Count (TPC) analysis time from 15–30 minutes to under 2 minutes per sample—an 85–90% increase in productivity, enabling laboratories to scale sample throughput without additional headcount.
- **Consistency & Reproducibility:** Eliminates inter-analyst variability (which typically ranges from 22.7%–80% in manual counting). By utilizing a standardized YOLOv8 AI pipeline, every test is processed with consistent logic, supporting ISO 17025 reproducibility requirements.
- **Cost Reduction:** Lowers operational labor costs by an estimated 40%. For high-volume facilities processing hundreds of samples daily, this efficiency results in significant annual savings in analyst labor hours.
- **Regulatory Compliance:** Provides an immutable, SHA-256 chained audit trail with mandatory analyst digital sign-offs, simplifying the path toward BPOM, SNI 2897:2008, and ISO 17025 accreditation.
- **Scientific Reliability:** Automated TNTC/TFTC flags and GUM measurement uncertainty (ISO/IEC Guide 98-3:2008) ensure every result is scientifically valid and defensible.
- **Democratization of Quality:** Enables smaller regional laboratories to maintain expert-level diagnostic accuracy, closing the quality gap across Indonesia's 500+ testing facilities.



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3.2 Shorts and Mid-Term Outcomes

Short-Term (0–6 Months):

- **Pilot Deployment:** Initiate implementation in 2–3 partner laboratories to validate $\geq 92\%$ detection accuracy across PCA, VRBA, and BGGB media types.
- **Capacity Building:** Onboard 10–20 analysts to the platform, utilizing embedded training materials to ensure seamless transition to the digital workflow.
- **Research Contribution:** Establish a labeled agar plate dataset of 1,477 images with 56,124 annotations. We will open-source this dataset to advance AI research in Indonesian food safety and microbiological diagnostics.

Mid-Term (6–24 Months):

- **Regional Scaling:** Expand deployment to 20+ accredited laboratories across Java and Sumatra, targeting 10,000+ monthly TPC analyses.
- **Ecosystem Integration:** Develop a mobile-native PWA for field data entry and achieve full API integration with existing Laboratory Information Management Systems (LIMS), such as SampleManager and LabVantage.
- **Sustainability:** Transition to a recurring SaaS subscription model, reaching operational sustainability within 18 months.
- **Validation & Publication:** Publish clinical validation study results in a peer-reviewed journal to secure scientific legitimacy and support national-level regulatory adoption.



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Innovation & Differentiation

4.1 What Makes Your Solution Different?

ColonyAI distinguishes itself from manual and high-cost commercial alternatives through several unique technical innovations:

- **5-Class Artifact Intelligence:** Unlike generic vision tools, our YOLOv8 model is specifically trained to classify 5 distinct objects (`colony_single`, `colony_merged`, `bubble`, `dust_debris`, `media_crack`), enabling >90% precision in artifact rejection.
- **SA-001 Merged Colony Estimation:** We go beyond simple bounding boxes by using area-ratio analysis to estimate counts within merged colonies, drastically improving accuracy compared to standard tools.
- **Metrological Traceability:** We are the only affordable platform calculating GUM-compliant measurement uncertainty ($k=2$) per ISO/IEC Guide 98-3:2008, essential for ISO 17025 accreditation.
- **Enterprise-Grade Security:** Our platform integrates SHA-256 chained immutable audit logs, 5-role RBAC, anti-phishing engines, and magic-bytes file validation—ensuring that diagnostic data is tamper-evident and secure.
- **No Hardware Lock-in:** By delivering as a browser-based SaaS, we eliminate the need for proprietary hardware costing USD 15,000–60,000, democratizing high-accuracy testing for regional laboratories.
- **Regulatory Context:** Built specifically for Indonesian laboratory standards, our automated PDF exports align perfectly with BPOM/SNI reporting requirements, addressing a critical gap left by international software.



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4.2 Positioning of the Solution Compared to Existing Approaches or Products

Feature	ColonyAI	Manual Counting	High-End Hardware	Generic AI API
5-Class Detection	✓	✗	Partial	✗
SA-001 Merged Estimation	✓	✗	✗	✗
GUM Uncertainty (ISO)	✓	✗	✗	✗
SHA-256 Audit Chain	✓	✗	✗	✗
Indonesian Regulatory Format	✓	✗	✗	✗
Cost	Affordable SaaS	High Labor	>\$15K-\$60K	Custom Dev

Strategic Advantage:

Manual counting is subjective and slow, while proprietary hardware (e.g., ProtoCOL 3) remains inaccessible for most regional laboratories due to high capital expenditure. Generic AI APIs lack domain-specific laboratory intelligence, such as artifact rejection and metrological uncertainty.

ColonyAI surpasses these by integrating advanced AI vision with regulatory-grade governance (audit logs, GUM uncertainty, and RBAC) in an affordable, hardware-free SaaS platform. We do not just count colonies; we provide an audit-ready "Laboratory Operating System" that empowers labs to meet ISO 17025 standards at a fraction of the cost, making high-precision microbiology testing accessible to every laboratory in Indonesia.



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Technical Approach

5.1 Main Solution

ColonyAI utilizes a modern, modular technology stack engineered for high-performance and regulatory compliance:

- **AI Vision Engine:** We use YOLOv8 for real-time 5-class detection, combined with OpenCV for image preprocessing (CLAHE normalization, Hough Circle for plate boundary extraction, and homography for perspective correction).
- **Backend & Data:** A FastAPI (Python) framework provides an asynchronous REST API that serves the inference pipeline and handles complex calculations, including SA-001 merged estimation and GUM-compliant uncertainty.
- **Frontend:** A Next.js 14 (TypeScript) dashboard delivers a high-speed, interactive user experience for plate annotation, historical trend analytics, and simulator benchmarking.
- **Security & Governance:** We employ PostgreSQL for ACID-compliant storage, featuring a SHA-256 chained audit log to ensure tamper-evident records. Application security is enforced through Argon2 password hashing, JWT blacklisting, and a 5-role RBAC model to strictly separate laboratory duties and ensure ISO 17025 compliance.

This stack enables an automated, scalable, and secure "Laboratory Operating System" accessible from any web browser.

5.2 Technology Selection and Implementation

We selected our technology stack to prioritize performance, security, and regulatory compliance:

- **YOLOv8:** Chosen for its superior balance of speed (inference in <50ms) and multi-class accuracy. Its ability to perform real-time detection without needing dedicated GPU hardware makes our platform accessible to standard laboratory PCs.
- **FastAPI:** Selected for its asynchronous capabilities, allowing for high-concurrency plate processing and auto-documented API endpoints that simplify integration with external laboratory systems (LIMS).
- **Next.js (TypeScript):** Provides a type-safe, performant frontend. Server-side rendering (SSR) ensures rapid dashboard loads, while React's component architecture allows us to build the interactive annotation tools essential for laboratory analysts.
- **PostgreSQL:** Chosen for strict ACID compliance. This is non-negotiable for our SHA-256 chained audit logs, which require robust data integrity to satisfy ISO 17025 accreditation.



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5.3 Solution Algorithm

ColonyAI leverages a high-performance, real-time pipeline:

- **AI Vision Engine:** We use a fine-tuned YOLOv8 model for multi-class object detection. Upon upload, the backend triggers an asynchronous inference process. Pre-processing uses OpenCV (CLAHE for normalization, Hough Circle Transform for plate boundary isolation) to standardize inputs. The model performs single-pass detection, outputting 5 classes (colony_single, colony_merged, bubble, dust_debris, media_crack) with confidence scores.
- **Merged Estimation (SA-001):** To ensure real-time accuracy on overlapping colonies, we implement an area-based heuristic. The system calculates the ratio of a merged bounding box area against the median single colony area to estimate the count, providing immediate results without complex, latency-heavy segmentation.
- **Cryptographic Integrity:** For regulatory compliance, we utilize a SHA-256 Chained Hash Algorithm. Every test record is cryptographically linked to the previous entry. This ensures that results are immutable and tamper-evident; any unauthorized modification invalidates the hash chain, satisfying ISO 17025 audit standards.

5.4 Primary Data or Input Used

ColonyAI relies on a high-integrity, domain-specific dataset curated for microbiology accuracy:

- **Custom-Labeled Dataset:** Our primary input consists of 1,477 real agar plate images captured in laboratory environments, containing 56,124 professional-grade bounding box annotations across our 5 detection classes. Every image was manually annotated and verified by domain experts to ensure clinical-grade labeling.
- **Data Augmentation:** To ensure robust real-time performance, we utilize the YOLOv8-native augmentation pipeline (Mosaic, HSV color-jittering, rotation, and scaling). This effectively multiplies our training data density, allowing the model to generalize across varied lighting, media types (PCA, VRBA, BGGB), and camera qualities.
- **Quality Controls:** We enforce strict data hygiene: minimum resolution (800x800px), double-blind annotation review, and class-balance verification. Plates with ≥ 300 CFU (TNTC) are excluded from training to maintain clear ground-truth targets.

This dataset provides the reliable foundation required for real-time inference. By training on diverse, real-world laboratory data, we ensure that our AI engine delivers stable and reproducible results, even under the inconsistent lighting or varied media types found in actual clinical laboratory settings.



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5.5 Security and Scalability Considerations

ColonyAI is built on an enterprise-grade architecture designed for both robust security and seamless scalability:

- **Data Security:** All sensitive images are encrypted at rest in S3-compatible storage with time-limited signed URLs. We enforce security-first ingestion: every file undergoes magic-bytes validation, ClamAV malware scanning, and EXIF metadata stripping to prevent malicious uploads and maintain user privacy.
- **Access Governance:** We implement a 5-role RBAC model, ensuring strict separation of laboratory duties. Access is protected by Argon2 password hashing and JWT blacklisting to prevent session hijacking.
- **Scalability:** The platform is containerized using Docker and deployed via Railway/Vercel, enabling horizontal auto-scaling to handle concurrent multi-laboratory demand. We utilize PgBouncer connection pooling to maintain database performance during heavy traffic.
- **Future Development:** Our CI/CD pipeline includes MLflow for model versioning, ensuring we can deploy improved YOLOv8 iterations without downtime. The modular architecture facilitates future LIMS integration and mobile-native PWA expansion.



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Implementation Feasibility

6.1 Invention Status

Current Stage: Prototype / Proof of Concept (POC)

ColonyAI has evolved beyond the theoretical stage into a fully functional technical prototype.

We have successfully implemented and verified the following core modules:

- **AI Vision Engine:** YOLOv8 model fine-tuned on 1,477 images, with a complete inference pipeline capable of real-time 5-class detection and SA-001 merged colony estimation.
- **Secure Backend:** FastAPI infrastructure is operational, featuring full image sanitization (magic-bytes, malware scanning, EXIF stripping), GUM-compliant calculations, and cryptographic SHA-256 audit logs.
- **Web Dashboard:** A responsive Next.js 14 application that supports image upload, annotated result visualization, historical data retrieval, and mandatory digital analyst sign-offs.
- **Security Stack:** Enterprise-grade security is live, including Argon2 password hashing, JWT blacklisting, account lockout, and a 5-role RBAC system that correctly enforces laboratory access control.
- **Database:** PostgreSQL schema is active, managing normalized user roles, detection results, and audit trails.

The platform is ready for pilot deployment in laboratory settings to collect final validation data and perform real-world reliability testing.

6.2 Is the Innovation Realistic to Build?

ColonyAI is highly realistic and strategically sound for our team for several key reasons:

- **Proven Tech Stack:** We leverage industry-standard frameworks (YOLOv8, FastAPI, Next.js) that have extensive community support, ensuring we can build, debug, and scale the system efficiently without reinventing the wheel.
- **Team Expertise:** Our members combine direct experience in computer vision, full-stack development, and microbiology workflow analysis, allowing for rapid parallel development across all system layers.
- **Operational Viability:** We utilize a cost-effective cloud architecture (Railway/Vercel) and free GPU resources (Google Colab) for model training, enabling enterprise-grade performance at near-zero infrastructure costs during the prototype phase.
- **Modular Design:** Because our system is built as a set of loosely coupled components (AI engine, API, dashboard), we can iterate, test, and improve each module independently.



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6.3 Development Stages

We utilize an Agile Scrum methodology with one-week sprints to ensure rapid, transparent development:

- Week 1 (Foundation): Dataset finalization and annotation QA (5-class taxonomy), YOLOv8 model fine-tuning, PostgreSQL database schema design, and FastAPI backend scaffolding.
- Week 2 (Core Platform): Integration of the inference pipeline with the Next.js dashboard, enabling real-time image upload, visual bounding-box annotation, and active Simulator module functionality.
- Week 3 (Integrations & Compliance): Development of the CFU/ml calculator with GUM measurement uncertainty, implementation of SHA-256 chained audit logs, and finalization of BPOM-compliant PDF report generation.
- Week 4 (QA & Launch): System-wide stress testing across all 5 classes, rigorous security audits (pen-testing the RBAC and auth flows), UI polish, and end-to-end documentation preparation for the pilot launch.

Each sprint concludes with a review and retrospective to ensure our "Definition of Done" is met, including code reviews, test coverage, and Product Owner sign-off. This disciplined approach ensures a seamless transition from prototype to a production-ready laboratory pilot.

6.4 Business Model and Sustainability

ColonyAI utilizes a SaaS (Software-as-a-Service) model, offering 30–40× ROI by replacing costly manual labor with automated high-precision diagnostics.

- **Customer Segments:** KAN-accredited labs, food manufacturers (QC departments), hospitals, and regulatory bodies (BPOM).
- **Value Proposition:** 85% faster TPC analysis, ISO 17025-compliant audit trails, and 5-class artifact intelligence at a fraction of the cost of \$15K+ proprietary hardware.
- **Revenue Streams:**
 - Starter (IDR 500K/mo): 500 analyses, ideal for smaller labs.
 - Professional (IDR 1.5M/mo): Unlimited usage and advanced analytics.
 - Enterprise: Custom LIMS integration, dedicated support, and SLA guarantees.
- **Cost Structure:** Cloud infrastructure (Railway/AWS), model retraining, continuous security compliance audits, and customer support.

Sustainability:

At IDR 500K/mo, a lab processing 200 samples daily saves ~IDR 15M–20M monthly in analyst time. We plan a 6-month free pilot program in 3 key laboratories to build high-quality validation data and customer testimonials. This data will refine our AI performance, creating a "flywheel effect" where higher accuracy leads to wider adoption. Revenue from Professional and Enterprise tiers will fund ongoing infrastructure, model improvements, and compliance certification, ensuring long-term operational sustainability.



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Attachment & Reference

A. System Visual Documentation:

ColonyAI System Architecture: <https://github.com/wi5nuu/colonyai/blob/main/project-management/technical-architecture.md>

Live Interface Demo: https://www.linkedin.com/posts/wisnu-alfian-nur-ashar_machinelearning-computervision-cybersecurity-ugcPost-7463508916144979968-

9_L_/?utm_source=share&utm_medium=member_desktop&rcm=ACoAAE8I924BbjE_sv16YWicBF3CwZCfAtgnK5M

Codebase Repository: <https://github.com/wi5nuu/colonyai>

B. Documentation:

User Manual: <https://github.com/wi5nuu/colonyai/blob/main/project-management/user-manual.md>

Security Features Checklist: https://github.com/wi5nuu/colonyai/blob/main/project-management/SECURITY_FEATURES_CHECKLIST.md

C. Citations & Standards:

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